



Inspiring Excellence

**CSE370 : Database Systems**  
**Project Report**

**Project Title : Animal Adoption Website**

| Group No: 03, CSE370 Lab Section: 01, Spring 2026 |                      |                                         |
|---------------------------------------------------|----------------------|-----------------------------------------|
| ID                                                | Name                 | Contribution                            |
| 23241004                                          | Afnan Mohammad Hafiz | Admin moderation,<br>Post search option |
| 23201336                                          | Nafiz Shahriar Sami  | Donation Campaign,<br>Donation Amount   |
| 23201393                                          | Eusha Kayenat        | Wishlist, Animal<br>Type                |

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## **Introduction**

The database website, called the Animal Welfare & Adoption Platform, known as Petverse, where it designed to bridge the gap between animal lovers, pet adopters, and welfare organizations. This is a sophisticated full-stack web application that integrates a centralized marketplace, a transparent donation system, and an adoption portal.

The platform is built on a modular architecture that addresses three key pillars of animal welfare: a centralized adoption portal, a transparent donation system, and an integrated moderation engine. The platform utilizes a modern Node.js and Express backend, coupled with a relational MySQL database. Key technical highlights include:

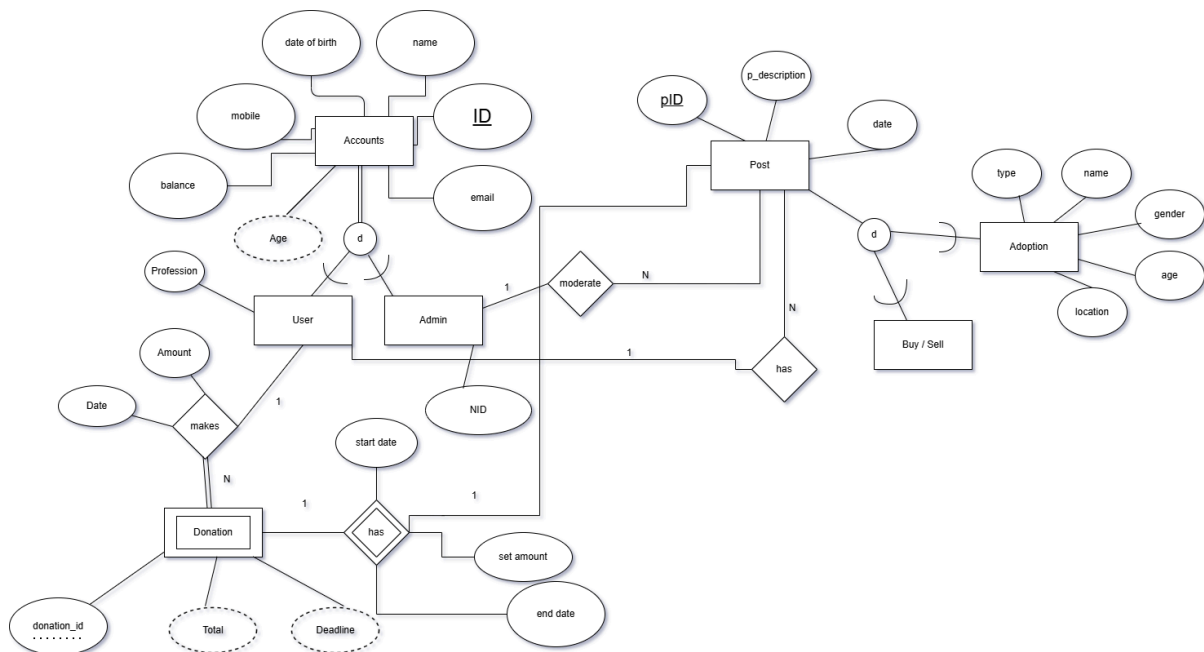
- **Secure Authentication:** Implementation of JWT (JSON Web Tokens) and bcrypt password hashing for user data protection.
- **Relational Integrity:** A complex database schema that manages supertype-subtype relationships (Accounts, Users, and Admins) and handles cascading data for adoptions and sales.
- **Transactional Stability:** Use of SQL transactions to ensure financial accuracy during donations and purchases.

Our goal is to ease the process of adopting pets and improve the lives of animals, creating a society that fosters connections between people and animals. By providing a secure, transparent, and user-friendly interface, the project facilitates higher adoption rates, provides a sustainable funding model for rescue operations, and builds a community dedicated to responsible pet ownership and welfare advocacy.

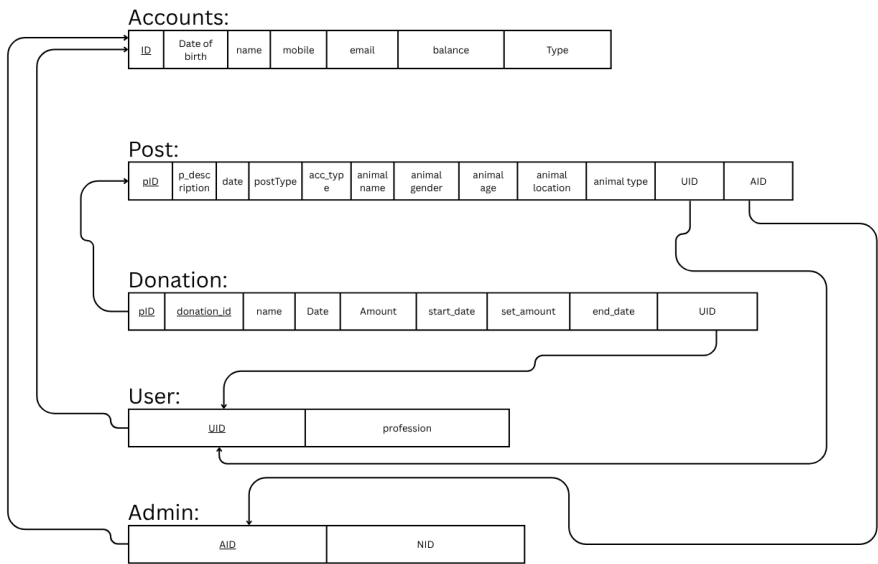
## Project Features

| ID, Name                              | Features [3 per member] |                                                                                                |
|---------------------------------------|-------------------------|------------------------------------------------------------------------------------------------|
| <b>23241004, Afnan Mohammad Hafiz</b> | Ft 1                    | Admin can moderate and delete posts                                                            |
|                                       | Ft 2                    | A post can be searched using any keywords of description, and acts similarly to that of CTRL+F |
| <b>23201336, Nafiz Shahriar Sami</b>  | Ft 1                    | Donation Amount Calculation and Time Handling                                                  |
|                                       | Ft 2                    | Donation Amount Track                                                                          |
| <b>23201393, Eusha Kayenat</b>        | Ft 1                    | Animal Type Dynamic Dropdown                                                                   |
|                                       | Ft 2                    | Wishlist                                                                                       |

## ER/EER Diagram



# Schema Diagram



## Normalization

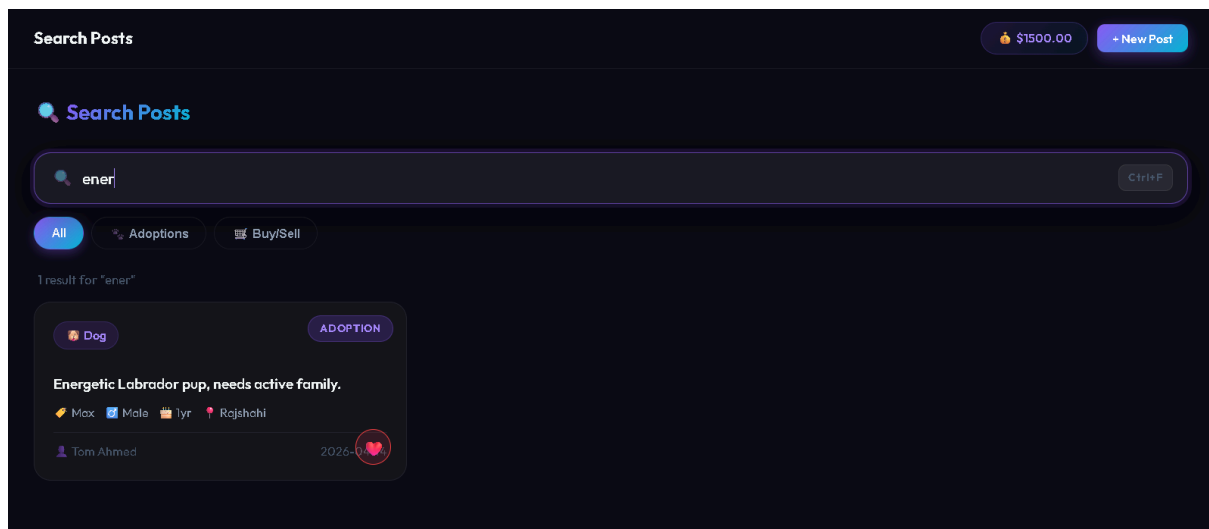
- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

- a. Yes, it is converted into 1NF.
- b. Yes, it is converted into 2NF. No, there cannot be any partial functional dependencies in our relational schema.
- c. Yes, it is converted into 3NF. No, there cannot be any transitive dependencies in our relational schema.

# Frontend Development

**Contribution of ID : 23241004, Name : Afnan Mohammad Hafiz**

The search bar is shown for all users so that they can fetch data with the keywords they want to see. The approval and deletion of posts made by the users can only be done by the admin. After approval of a post, a user can see the post in their feed, and it is also added to the moderated list of the admin. Moreover, there is also a dashboard showing total counts for many sections.



The screenshot shows an 'All Posts' interface. At the top right, there's an 'Admin Mode' button. Below the header, there are tabs for 'All', 'Adoptions', and 'Buy/Sell'. The 'All' tab is selected. Below the tabs, there's a table with the following columns: ID, AUTHOR, TYPE, DETAILS, DATE, STATUS, and ACTIONS.

| ID  | AUTHOR                        | TYPE     | DETAILS                             | DATE       | STATUS    | ACTIONS                                                      |
|-----|-------------------------------|----------|-------------------------------------|------------|-----------|--------------------------------------------------------------|
| #20 | Joseph<br>jose@bracu.com      | Adoption | Bear - Long                         | 2026-05-03 | Pending   | <input checked="" type="checkbox"/> <input type="checkbox"/> |
| #17 | Sara Khan<br>sara@example.com | Adoption | Cat - Chiki                         | 2026-04-28 | Pending   | <input checked="" type="checkbox"/> <input type="checkbox"/> |
| #9  | Rina Mou<br>rina@example.com  | Buy/Sell | Medicine - \$0.00 - Exp: 2027-03-14 | 2026-04-14 | Pending   | <input checked="" type="checkbox"/> <input type="checkbox"/> |
| #8  | Tom Ahmed<br>tom@example.com  | Buy/Sell | Toys - \$0.00                       | 2026-04-14 | Pending   | <input checked="" type="checkbox"/> <input type="checkbox"/> |
| #7  | Sara Khan<br>sara@example.com | Buy/Sell | Medicine - \$0.00 - Exp: 2026-12-30 | 2026-04-14 | Moderated | <input checked="" type="checkbox"/> <input type="checkbox"/> |
| #6  | Rina Mou<br>rina@example.com  | Adoption | Bird - Polly - Khulna               | 2026-04-14 | Moderated | <input checked="" type="checkbox"/> <input type="checkbox"/> |
| #5  | Tom Ahmed<br>tom@example.com  | Adoption | Dog - Max - Rajshahi                | 2026-04-14 | Moderated | <input checked="" type="checkbox"/> <input type="checkbox"/> |

Colorful parrot, can say 10+ words!

ADOPTION

 Bird

 Polly

 Khulna

 Rina Mou <rina@example.com>

#6

 Moderated

 Delete

Admin Overview

Admin Mode



10

Total Posts



7

Adoptions



7

Users



4

Moderated



9

Deleted



\$54303

Donations Collected

Recent Posts



**Contribution of ID : 23201336, Name : Nafiz Shahriar Sami**

The donations section has donation campaigns, and each campaign has a list of features. Such as how many days are left (calculated from start date and end date), the amount of money raised (calculated from the donations made by different users), the target amount as a goal, active status (based on the days left feature), and the set amount.

## Dog Shelter Fund

Active

 Dog

Building a new shelter for rescued dogs in Chittagong.

Raised: \$32152.68

Goal: \$80000.00

40.2% funded

 2026-02-28 → 2026-09-29 •  147 days left

Amount (\$)

 Donate

## New Campaign

### + New Campaign

CAMPAIGN TITLE

e.g. Save Street Cats

DESCRIPTION

What is this campaign for?

START DATE

mm/dd/yyyy



END DATE

mm/dd/yyyy



TARGET AMOUNT (\$)

e.g. 5000

ANIMAL TYPE (OPTIONAL)

None (General)

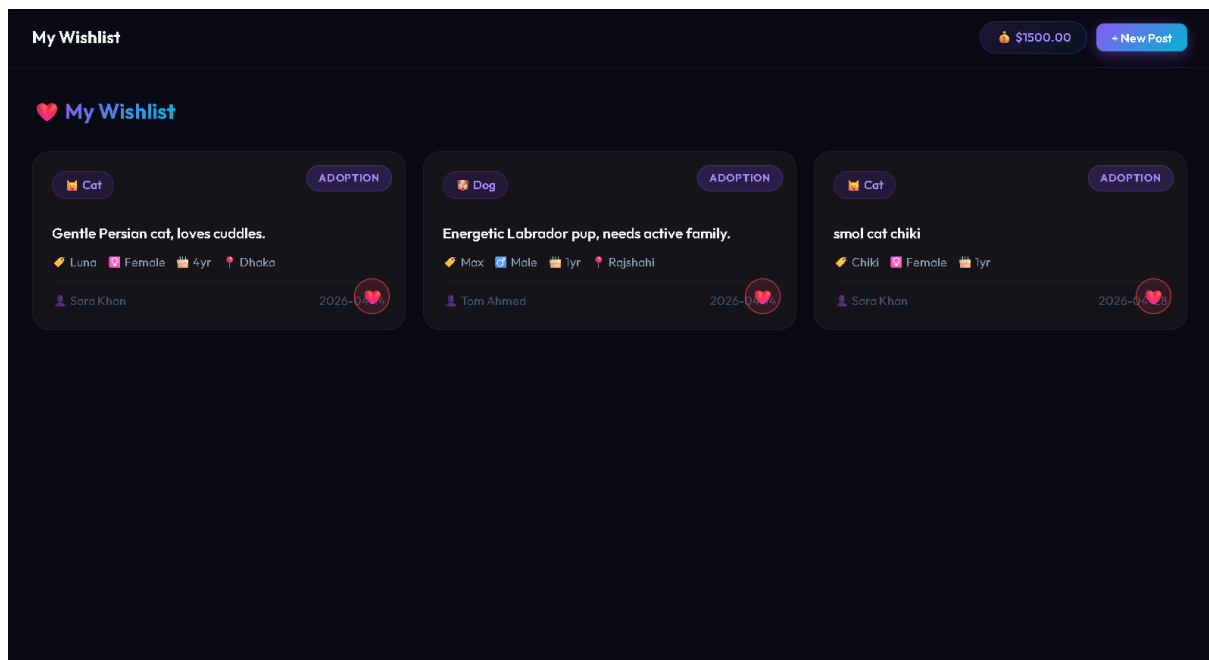


Launch Campaign



**Contribution of ID : 23201393, Name : Eusha Kayenat**

**The wishlist feature on our website works when we click on the heart button in the bottom right corner of the Adoption and Buy/Sell posts from the dashboard. The heart remains colorless until it has been clicked. And whenever the user clicks the heart icon, it turns red. The other feature that I have worked on is animal type. Whenever the user clicks on the ‘Animal Type’ drop-down bar, it shows all the animal names that are available in the posts. The user can select any one of the animals listed on it, and all posts related to that animal will be displayed on the screen.**



## Adoptions

### Adoption Posts

All Animal Types 

All Animal Types

 Bear

 Bird

 Cat

 Dog

 Snake

ADOPTION

say 10+ words!

5yr  Khulna

 Rina Mou

2026-04-14 

## Backend Development

**Contribution of ID : 23241004, Name : Afnan Mohammad Hafiz**

The admin, being a part of the account with different permissions and authentication access, can moderate and delete posts made by the users. After the admin moderates the posts, the post will show on the website for all the users to see and will be added to the list of moderated posts. The same will happen for the case of deletion of a post. Moreover, the search option in the post will fetch only the desired keyword-related posts from the approved posts made by the admin. This will serve as the function of CTRL+F in the website for post fetching by the users.

```
// -----  
// POST /api/admin/moderate/:postId - mark a post as moderated  
// -----  
router.post('/moderate/:postId', authenticate, requireAdmin, async (req, res) => {  
  const { postId } = req.params;  
  const { notes } = req.body;  
  try {  
    // Check post exists  
    const [postRows] = await pool.query('SELECT post_id FROM Posts WHERE post_id = ?', [postId]);  
    if (postRows.length === 0) return res.status(404).json({ message: 'Post not found' });  
  
    // INSERT IGNORE so re-moderating same post by same admin is ignored  
    await pool.query(  
      'INSERT IGNORE INTO Moderation (admin_id, post_id, notes) VALUES (?, ?, ?)',  
      [req.user.account_id, postId, notes || null]  
    );  
    res.json({ message: 'Post marked as moderated' });  
  } catch (err) {  
    console.error('Admin moderate error:', err);  
    res.status(500).json({ message: 'Server error' });  
  }  
});
```

```

// -----
// DELETE /api/admin/posts/:postId - admin deletes a post
// -----
router.delete('/posts/:postId', authenticate, requireAdmin, async (req, res) => {
  const post_id = parseInt(req.params.postId);
  const conn = await pool.getConnection();
  try {
    await conn.beginTransaction();

    // Verify post exists
    const [postRows] = await conn.query('SELECT post_type FROM Posts WHERE post_id = ?', [post_id]);
    if (postRows.length === 0) {
      await conn.rollback();
      return res.status(404).json({ message: 'Post not found' });
    }

    // Get animal_type_id before cascade delete
    let animal_type_id = null;
    if (postRows[0].post_type === 'Adoption') {
      const [adRows] = await conn.query('SELECT animal_type_id FROM Adoptions WHERE post_id = ?', [post_id]);
      if (adRows.length > 0) animal_type_id = adRows[0].animal_type_id;
    }

    // Record deletion in Delete_Post log
    await conn.query(
      'INSERT INTO Delete_Post (admin_id, post_id) VALUES (?, ?)',
      [req.user.account_id, post_id]
    );

    // Delete post (cascades to Adoptions, BuySell, Medicine, Toys, Moderation)
    await conn.query('DELETE FROM Posts WHERE post_id = ?', [post_id]);
  } catch (err) {
    await conn.rollback();
    return res.status(500).json({ message: 'Database error' });
  }
});

```

```

    params.push(category);
  }
  if (search && search.trim()) {
    sql += ' AND p.description LIKE ?';
    params.push(`%${search.trim()}%`);
  }
}

```

## Contribution of ID : 23201336, Name : Nafiz Shahriar Sami

A donation campaign is something only admins can post, and only users can donate to the said campaign. The campaigns require a start date and an end date, and the number of days left is calculated and shown to the users. When a user wants to donate, they need to set the amount they want to donate. Afterwards, that amount will be deducted from the user's balance and added to the campaign. The added value will stack, and the percentage of donation in that campaign will be shown.

```
router.get('/:id', async (req, res) => {
  try {
    const [rows] = await pool.query(
      `SELECT d.*, at.name AS animal_type_name,
        ROUND((d.current_amount / d.target_amount) * 100, 1) AS progress_pct
      FROM Donations d
      LEFT JOIN AnimalTypes at ON at.animal_type_id = d.animal_type_id
      WHERE d.donation_id = ?`,
      [req.params.id]
    );
    if (rows.length === 0) return res.status(404).json({ message: 'Campaign not found' });
    res.json(rows[0]);
  } catch (err) {
    res.status(500).json({ message: 'Server error' });
  }
});
```

```
router.post('/:id/contribute', authenticate, async (req, res) => {
  if (req.user.role !== 'user') {
    return res.status(403).json({ message: 'Only users can donate' });
  }

  const donation_id = parseInt(req.params.id);
  const amount = parseFloat(req.body.amount);

  if (!amount || amount <= 0) {
    return res.status(400).json({ message: 'amount must be a positive number' });
  }

  const conn = await pool.getConnection();
  try {
    await conn.beginTransaction();

    // Lock campaign row
    const [campaigns] = await conn.query(
      'SELECT donation_id, end_date, target_amount, current_amount FROM Donations WHERE donation_id = ? FOR UPDATE',
      [donation_id]
    );
    if (campaigns.length === 0) {
      await conn.rollback();
      return res.status(404).json({ message: 'Campaign not found' });
    }
    const campaign = campaigns[0];

    // Check campaign still active
    const today = new Date().toISOString().slice(0, 10);
    if (today > campaign.end_date) {
      await conn.rollback();
      return res.status(400).json({ message: 'This campaign has ended' });
    }
  }
});
```



```

// Deduct from user balance
await conn.query(
  'UPDATE Accounts SET balance = balance - ? WHERE account_id = ?',
  [amount, req.user.account_id]
);

// Increase campaign amount
await conn.query(
  'UPDATE Donations SET current_amount = current_amount + ? WHERE donation_id = ?',
  [amount, donation_id]
);

// Record transaction
await conn.query(
  'INSERT INTO Donation_Transactions (user_id, donation_id, amount, date) VALUES (?, ?, ?, ?)',
  [req.user.account_id, donation_id, amount, today]
);

await conn.commit();

// Return updated balance
const [updatedAcc] = await pool.query('SELECT balance FROM Accounts WHERE account_id = ?', [req.user.account_id]);
res.json({ message: 'Donation successful!', new_balance: updatedAcc[0].balance });
} catch (err) {
  await conn.rollback();
  console.error('Contribute error:', err);
  res.status(500).json({ message: 'Donation failed' });
} finally {
  conn.release();
}

```

## Contribution of ID : 23201393, Name : Eusha kayenat

The wishlist is a feature where, on the bottom right corner of every post, there is a heart icon; clicking it changes the icon's color to red and takes the post into the wishlist section. The wishlist section is user-specific, meaning that each user has their own wishlist. Clicking on the heart icon again removes that specific post from the wishlist. The adoption post type has a drop-down bar that indicates animal types. It is a dynamic drop-down bar, meaning that new animal types will be added here as long as there is one post about that animal, and if an animal in the list has no posts related to it, it will be removed from the list.

```
// POST /api/wishlist/:postId - Add to wishlist
router.post('/:postId', authenticate, requireUser, async (req, res) => {
  const { postId } = req.params;
  const userId = req.user.account_id;

  try {
    await pool.query(
      'INSERT IGNORE INTO Wishlists (user_id, post_id) VALUES (?, ?)',
      [userId, postId]
    );
    res.status(201).json({ message: 'Added to wishlist' });
  } catch (err) {
    console.error('Add to wishlist error:', err);
    res.status(500).json({ message: 'Failed to add to wishlist' });
  }
});

// DELETE /api/wishlist/:postId - Remove from wishlist
router.delete('/:postId', authenticate, requireUser, async (req, res) => {
  const { postId } = req.params;
  const userId = req.user.account_id;

  try {
    await pool.query(
      'DELETE FROM Wishlists WHERE user_id = ? AND post_id = ?',
      [userId, postId]
    );
    res.json({ message: 'Removed from wishlist' });
  }
});
```

```

try {
  // The query is similar to GET /api/posts but filtered by Wishlists table
  const query = `
    SELECT p.post_id, p.description, p.post_date, p.post_type,
           a.name AS author_name,
           m.post_id AS moderated,
           ad.animal_name, ad.gender, ad.age, ad.location,
           at.name AS animal_type_name,
           bs.category, bs.price,
           med.expire_date,
           1 AS is_wishlisted
    FROM Wishlists w
    JOIN Posts p ON w.post_id = p.post_id
    JOIN Accounts a ON p.account_id = a.account_id
    LEFT JOIN Moderation m ON p.post_id = m.post_id
    LEFT JOIN Adoptions ad ON p.post_id = ad.post_id
    LEFT JOIN AnimalTypes at ON ad.animal_type_id = at.animal_type_id
    LEFT JOIN BuySell bs ON p.post_id = bs.post_id
    LEFT JOIN Medicine med ON bs.buysell_id = med.buysell_id
    WHERE w.user_id = ?
    AND NOT EXISTS (SELECT 1 FROM Delete_Post dp WHERE dp.post_id = p.post_id)
    ORDER BY w.created_at DESC
  `;
  const [posts] = await pool.query(query, [userId]);
  res.json({ posts });
}

```

```

1  const router = require('express').Router();
2  const pool = require('../db');
3
4  // GET /api/animal-types
5  // Returns all animal types currently in the AnimalTypes table
6  router.get('/', async (req, res) => {
7    try {
8      const [rows] = await pool.query('SELECT * FROM AnimalTypes ORDER BY name ASC');
9      res.json(rows);
10   } catch (err) {
11     console.error('AnimalTypes error:', err);
12     res.status(500).json({ message: 'Server error' });
13   }
14 });
15
16 module.exports = router;
17

```

## Source Code Repository

[Google Drive Link](#)

## Conclusion

In conclusion, Petverse demonstrates how a well-designed, full-stack platform can meaningfully transform the landscape of animal welfare by unifying adoption, donation, and community engagement into a single, secure ecosystem. By leveraging modern technologies and thoughtful design, Petverse creates a sustainable and scalable solution that supports rescue efforts, encourages responsible pet ownership, and strengthens the bond between humans and animals. Ultimately, the platform stands as a practical step toward a more compassionate and connected society where animal welfare is both accessible and actively supported.

## References

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<https://www.youtube.com/watch?v=f-r4E968-3E>

Lama Dev. (n.d.). *Node.js Express MySQL full stack project* [Video]. YouTube.  
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